

CHAPTER 4: HARDWARE COMPONENTS DESCRIPTION

4.1 Introduction

Hardware components are the backbone of any embedded and IoT system. The performance, reliability, and efficiency of a Smart Irrigation System depend on the proper selection and integration of hardware devices.

The proposed IoT Smart Irrigation and Plant Monitoring System consists of the following hardware components:

1. ESP32 Development Board
2. DHT11 Temperature and Humidity Sensor
3. Soil Moisture Sensor
4. OLED Display (128×64 I2C)
5. Relay Module
6. Mini Water Pump
7. Breadboard
8. Jumper Wires
9. USB Cable

Each component plays an important role in monitoring environmental conditions and controlling irrigation.

4.2 ESP32 Development Board

4.2.1 Introduction

ESP32 is a low-cost, high-performance microcontroller developed by Espressif Systems. It is widely used in Internet of Things (IoT) applications because it integrates Wi-Fi, Bluetooth, processing power, and GPIO capabilities on a single chip.

In this project, ESP32 acts as the central controller responsible for acquiring sensor data, processing information, displaying results, and controlling irrigation.

4.2.2 Features of ESP32

- Dual-Core Processor
- Clock Speed up to 240 MHz
- Built-in Wi-Fi

- Built-in Bluetooth
 - Multiple GPIO Pins
 - ADC Channels
 - PWM Support
 - Low Power Consumption
 - Flash Memory Support
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4.2.3 Technical Specifications

Parameter	Value
Operating Voltage	3.3V
CPU	Dual Core Xtensa
Clock Speed	Up to 240 MHz
ADC Resolution	12-bit
Wi-Fi	802.11 b/g/n
Bluetooth	BLE + Classic
GPIO Pins	34+

4.2.4 Role in Project

The ESP32 performs the following tasks:

- Reads DHT11 sensor data.
 - Reads soil moisture values.
 - Calculates moisture percentage.
 - Updates OLED display.
 - Controls relay module.
 - Executes irrigation algorithm.
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4.3 DHT11 Temperature and Humidity Sensor

4.3.1 Introduction

The DHT11 is a digital sensor used to measure ambient temperature and humidity.

It contains:

- Humidity sensing element
- Temperature sensing element
- Internal signal processing unit

The sensor outputs calibrated digital data which can be directly read by the ESP32.

4.3.2 Features

- Low Cost
 - Digital Output
 - Easy Interfacing
 - Reliable Performance
 - Factory Calibration
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4.3.3 Technical Specifications

Parameter	Value
Operating Voltage	3.3V-5V
Temperature Range	0°C-50°C
Humidity Range	20%-90% RH
Accuracy	±2°C
Interface	Digital

4.3.4 Pin Description

Pin	Function
VCC	Power Supply
DATA	Data Output
GND	Ground

4.3.5 Working Principle

The DHT11 continuously measures temperature and humidity and converts the readings into digital signals.

The ESP32 receives this data through GPIO4 and processes it for display and monitoring.

4.4 Soil Moisture Sensor

4.4.1 Introduction

The Soil Moisture Sensor is used to determine the water content present in soil.

It is one of the most important sensors in irrigation systems because irrigation decisions are based on soil moisture levels.

4.4.2 Working Principle

The sensor measures soil conductivity.

- Wet Soil → Higher Conductivity
- Dry Soil → Lower Conductivity

The sensor produces analog voltage proportional to moisture content.

ESP32 converts this analog voltage into digital values using its ADC.

4.4.3 Features

- Simple Operation
 - Analog Output
 - Low Cost
 - Easy Integration
 - Fast Response
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4.4.4 Technical Specifications

Parameter	Value
Voltage	3.3V-5V
Output Type	Analog
Response Time	Fast
Power Consumption	Low

4.4.5 Calibration

The following calibration values were used:

AIR_VALUE = 4095

WATER_VALUE = 1200

These values help convert raw ADC values into moisture percentages.

4.4.6 Role in Project

The sensor continuously monitors soil moisture and provides data to ESP32 for irrigation decisions.

4.5 OLED Display (128×64 I2C)

4.5.1 Introduction

OLED (Organic Light Emitting Diode) displays are widely used in embedded systems because of their low power consumption and excellent visibility.

The OLED display used in this project provides real-time information to users.

4.5.2 Features

- High Contrast
 - Low Power Consumption
 - Compact Size
 - I2C Communication
 - Wide Viewing Angle
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4.5.3 Technical Specifications

Parameter	Value
Resolution	128 × 64
Communication	I2C

Parameter	Value
Voltage	3.3V-5V
Display Type	Monochrome OLED

4.5.4 Pin Description

OLED Pin	ESP32 Pin
VCC	3.3V
GND	GND
SDA	GPIO21
SCL	GPIO22

4.5.5 Information Displayed

The OLED screen displays:

- Temperature
- Humidity
- Soil Moisture Percentage
- Pump Status

4.5.6 Advantages

- No Backlight Required
 - Excellent Visibility
 - Fast Refresh Rate
 - Compact Design
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4.6 Relay Module

4.6.1 Introduction

A relay is an electrically operated switch used to control high-power devices using low-power signals.

The relay module allows ESP32 to control the water pump safely.

4.6.2 Relay Used

Relay Model:

JQC3F-05VDC-C

Relay Type:

5V Active LOW Relay Module

4.6.3 Features

- Electrical Isolation
 - High Reliability
 - Easy Interfacing
 - LED Status Indicator
 - Low Power Control
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4.6.4 Technical Specifications

Parameter	Value
Coil Voltage	5V
Control Type	Active LOW
Contact Type	COM, NO, NC
Switching Current	Up to 10A

4.6.5 Working Principle

When ESP32 sends LOW signal to GPIO26:

- Relay activates.
- Contacts switch.
- Pump turns ON.

When ESP32 sends HIGH signal:

- Relay deactivates.
- Pump turns OFF.

4.6.6 Relay Connections

Relay Pin	Connection
VCC	VIN (5V)
GND	GND
IN	GPIO26

4.7 Mini Water Pump

4.7.1 Introduction

The water pump is the actuator responsible for supplying water to plants.

It receives power through the relay module.

4.7.2 Features

- Compact Size
- Low Voltage Operation
- Low Power Consumption
- Suitable for Mini Irrigation Projects

4.7.3 Technical Specifications

Parameter	Value
Voltage Range	3V–9V
Current Consumption	Low
Flow Type	DC Pump

4.7.4 Working Principle

When the relay activates, electrical power reaches the pump.

The pump draws water and supplies it to plants.

When the relay deactivates, the pump stops.

4.8 Breadboard

4.8.1 Introduction

A breadboard is used for temporary circuit construction without soldering.

It simplifies testing and debugging.

4.8.2 Advantages

- Reusable
 - No Soldering Required
 - Easy Modifications
 - Fast Prototyping
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4.8.3 Power Distribution

Current project configuration:

Left Red Rail → 3.3V

Left Blue Rail → GND

Right Red Rail → VIN (5V)

Right Blue Rail → GND

Both GND rails are interconnected.

4.9 Jumper Wires

Jumper wires are used to establish electrical connections between various components.

Types:

- Male-to-Male
- Male-to-Female
- Female-to-Female

They provide flexibility during prototyping.

4.10 USB Cable

The USB cable serves two purposes:

1. Power Supply
2. Program Uploading

The cable connects ESP32 to the computer for firmware development and testing.

4.11 Hardware Integration Summary

The integration of all hardware components forms a complete smart irrigation system.

The sensors continuously provide environmental data to ESP32. The controller processes the information and displays it on the OLED screen. Based on moisture levels, the relay activates or deactivates the water pump, enabling automated irrigation.

The selected hardware components provide a low-cost, reliable, and scalable solution suitable for smart agriculture applications.

4.12 Summary

This chapter explained the hardware components used in the IoT Smart Irrigation and Plant Monitoring System. Detailed descriptions, specifications, working principles, and project roles of each component were presented. The hardware setup forms the foundation of the complete system and enables efficient monitoring and automated irrigation functionality.